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SprayCut Agro Rover

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ABSTRACT

Agriculture continues to be a cornerstone of India's economic structure. and pesticide spraying plays a crucial role in protecting crops from harmful insects. The proposed system is a versatile agricultural robot that integrates grass cutting with automated pesticide spraying, providing an efficient and smart solution for modern farming. The system supports remote operation via a mobile interface, allowing farmers to supervise and control the robot safely from a distance.

The main objective of this project is to develop an real agricultural robot integrated with a grass-cutting mechanism that optimizes pesticide usage, minimizes human exposure to harmful chemicals, and reduces labour requirements. The robot is equipped with a cutting mechanism, driving control, a spraying mechanism, and obstacle avoidance using a multi sensor module. Simulations and analyses are performed for spraying, cutting and sensor integration to enhance system performance. The proposed system significantly decreases manual labour requirements, limits environmental contamination, enhances agricultural productivity, and ensures improved safety for farmers and field workers.

Keywords: Intelligent pesticide spraying, Grass cutting mechanism, Multifunctional farming robot, Sensor-based obstacle avoidance, Mobile-operated agricultural system

I. INTRODUCTION

In India, nearly about 60% people are depending upon agriculture. As compared to other fields globalization in agriculture system is less. So, it is necessary to make some advancement in this field [1]. Robotics for agricultural purposes can aid the execution of repetitive and heavy manual operations improving, the conditions of agricultural workers [2]. To maintain yields, crops require regular fertilizer and pesticide application. Manual fertilizer and pesticide application by workers with knapsack sprayers are methods of the past which are not only highly inefficient and time-consuming, but also result in high labour requirements and costs to cover large agriculture fields [3]. The productivity of crop is affected by other major biological parameter such as pest, disease and this parameter can be control by human being for improving production of crop. But it is very harmful procedure for farmers when they spray pesticide, they have to take too many precautions like wearing suitable outfit, gloves and masks etc. This system is based on developing a robotic vehicle employed in agriculture for spraying dangerous pesticides [4].

To address these challenges, technological innovation in agriculture has become essential. Robotic systems offer a promising solution to improve productivity, efficiency, and safety. The development of an autonomous agricultural pesticide spraying robot provides an intelligent and cost-effective alternative to traditional methods [5]. Such a robot ensures uniform pesticide distribution, reduces chemical wastage, minimize human exposure to toxic substances, and lowers labour dependency. In this project, a robotic vehicle is designed for agricultural pesticide spraying using an Arduino micro controller. The robot is remotely operated through bile application, enabling farmers to control movement and spraying operations from a safe distance. This affordable system enhances safety, increases crop protection efficiency, and supports modernizing agricultural practices to meet growing labour and food demands [6]

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II. METHODOLOGY

Block Diagram

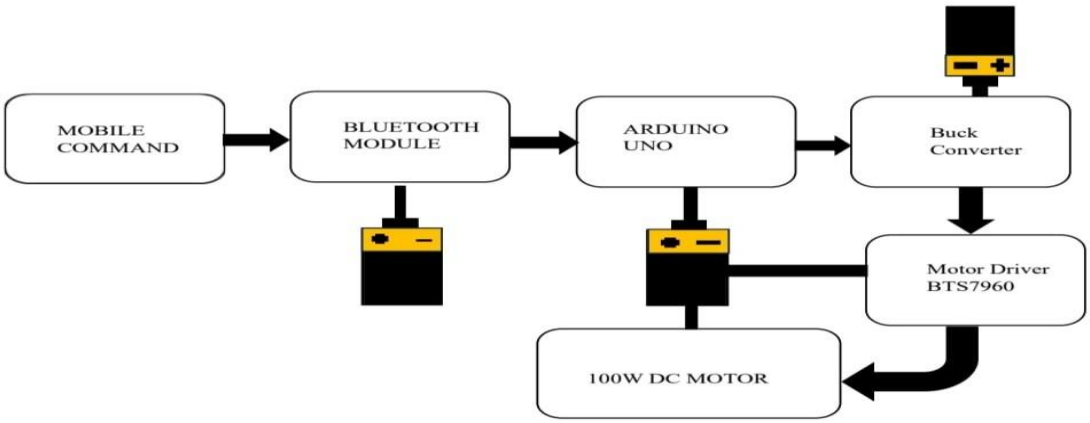


Figure 1. Block Diagram

The diagram represents an Arduino-based Bluetooth controlled DC motor system used in applications such as a pesticide spraying and grass-cutting robot. In this system, commands are sent from a mobile phone through a Bluetooth application to a Bluetooth module (such as HC-05/HC-06), which wirelessly receives the signals and forwards them to the Arduino Uno. The Arduino Uno acts as the main controller, processing the received commands and generating appropriate control signals. These signals are sent to the BTS7960 motor driver, which is capable of handling high current and controlling the direction and speed of the 100W DC motor using PWM. A buck converter is used to step down the battery voltage to the required level to safely power the Arduino and other components. The battery serves as the main power source for the entire system. Overall, the system enables wireless control of a high-power DC motor, making it suitable for automated agricultural applications.

Major components

A. Arduino Uno: The Arduino Uno functions as the main control unit of the robot. It processes the commands received through the Bluetooth module and generates appropriate control signals for the motor driver. By managing motor speed and direction, it ensures accurate and smooth movement of the robot.
B. BTS7960 Motor Driver: The BTS7960 motor driver acts as an interface between the Arduino and the DC motors. It amplifies low-power control signals from the Arduino to drive high-power motors. The driver supports forward and reverse motion along with speed regulation and can safely handle high current loads.
C. HC-05 Bluetooth Module: The HC-05 Bluetooth module enables wireless communication between the robot and a smartphone. It transfers user commands to the Arduino, allowing the robot to be operated remotely without the need for wired connections.
D. Buck Converter (Step-Down Module): The buck converter is used to step down the battery voltage to a regulated 5V supply required by the Arduino and other electronic components. It helps prevent voltage fluctuations and protects sensitive circuits, ensuring reliable system operation.

III. MODELLING AND ANALYSIS

The modelling of the proposed system focuses on the design and structure of the Arduino-based pesticide spraying robot with grass cutter. It includes system architecture, mechanical design, circuit design, and software flow.

1.1 System Architecture

The system consists of an Arduino controller, motor driver, DC motors for movement, a water pump for pesticide spraying, a cutting motor for grass cutting, and a rechargeable battery. All components are

interconnected to perform automated agricultural operations. The Arduino acts as the brain of the system and controls all actions based on programmed instructions.

1.2 Circuit Design

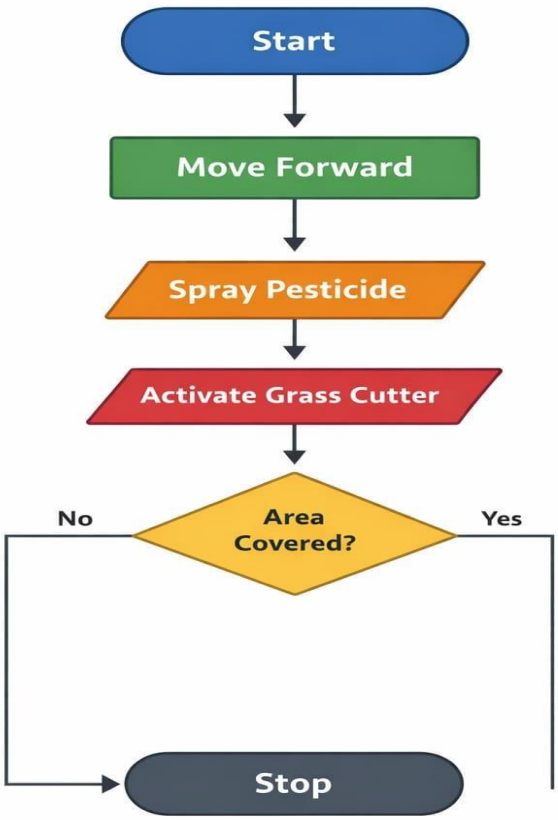
The circuit consists of Arduino connected to a motor driver module which controls the DC motors. The pump and cutting motor are also interfaced with Arduino through suitable drivers or relays. The battery provides power to all components.

1.3 Mechanical Design

The robot is designed with a wheeled base for mobility. The grass cutter is mounted at the bottom to cut grass efficiently, while the spray nozzle is positioned above to ensure uniform pesticide distribution. The structure is compact and suitable for agricultural fields

1.4 Working Analysis

The robot moves across the field using DC motors. Simultaneously, the pesticide pump sprays liquid through the nozzle, and the grass cutter trims unwanted grass. The operations are synchronized to ensure effective farming.



Flowchart of Pesticide Spraying & Grass Cutting Robot

IV. RESULTS AND DISCUSSION

The developed robot successfully performed all the intended. functions, including smooth forward and backward movement, wireless control through a Bluetooth interface, and stable. operation of high-power motors using the BTS7960 driver. The 12V 25Ah battery provided sufficient current to the robot continuously, and the metal chassis offered strong support on uneven agricultural surfaces. The robot was able to carry additional load such the pesticide tank or grass cutting mechanism without losing stability. Overall, the system demonstrated reliable performance, accurate control response, and effective mobility, fulfilling the objectives of designing a low-cost and efficient agricultural assistance robot.

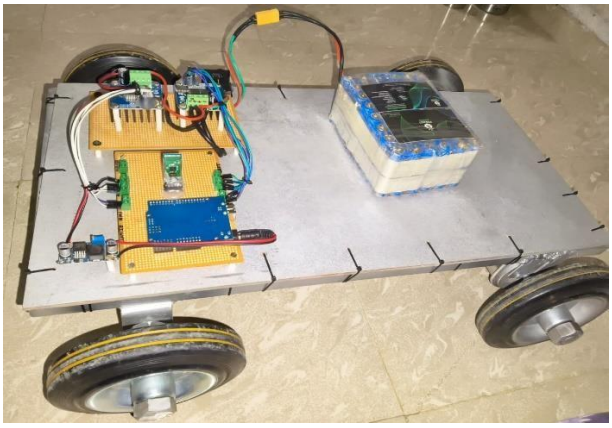


Figure 2. Image of Project

V. CONCLUSION

The pesticide robot with cutter innovative step towards modernizing agriculture practices through automation. By combining sensors and automation technology helps farmers save time, reduce labour cost and minimize health risk by limiting their exposure to harmful chemicals and improve operational efficiency. The use of sensors and microcontrollers ensures precision and reliability in field operations. Overall, this robot provides a cost-effective, eco-friendly, and time-saving solution that supports sustainable farming and enhances the productivity of farmers.

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